NeuroPriv: Adversarial Representation Learning for Privacy in Wearable EEG Systems

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Abstract

Wearable EEG systems may expose sensitive information beyond their intended health function, creating substantial risks to neuroprivacy. In this work, we show that commonly used EEG features can reveal participant identity and demographic attributes in addition to supporting the intended cognitive task. Wearable EEG is increasingly being explored for cognitive monitoring, neurological assessment, and longitudinal digital-health applications, yet many systems assume that transmitting compact spectral or spatial features instead of raw EEG provides sufficient privacy protection. Using EEGMAT as a motivating case study, we find that compact EEG features achieve a balanced accuracy of 0.788 for cognitive-state classification while enabling gender, age, and subject-identity inference with balanced accuracies of 0.858, 0.789, and 0.692, respectively. We further show that privacy-aware representation learning preserves task performance at 0.781 while reducing these inference accuracies to 0.563, 0.467, and 0.206. These findings motivate purpose-limited representations and explicit privacy auditing in wearable neurohealth systems.

Keywords

EEG, wearable sensing, neuroprivacy, privacy-preserving machine learning, adversarial learning

1 Introduction

Wearable EEG sensing is moving beyond clinical and laboratory settings into everyday computing [9, 16]. Emerging head-worn systems support applications such as cognitive-state monitoring, workload estimation, brain–computer interaction, sleep analysis, and mobile health assessment [3, 5, 7, 13]. To make these applications practical on resource-constrained devices, wearable EEG pipelines commonly replace raw-signal transmission with compact spectral and spatial features that can be processed efficiently by downstream machine learning models. This low-latency design remains privacy-sensitive. Traditional EEG features are often treated as safer than raw signals because they remove waveform-level detail and explicit identifiers. However, removing raw data does not necessarily remove identifiability [6]. EEG contains subject-specific and demographic information, and prior work has shown that users can be identified from EEG-derived representations even when direct identifiers are absent [14]. In this work, we show that a wearable system may avoid releasing raw EEG while still exposing sensitive information through the features it transmits.

We examine this problem on EEGMAT [21], where the intended utility task is Baseline-versus-Arithmetic classification and the private attributes are gender, age, and subject identity.Our analysis shows that the same feature representation used for task inference also supports strong private-attribute inference. The best task classifier achieves a balanced accuracy of 0.788, while gender, age, and subject identity reach 0.858, 0.789, and 0.692, respectively. These results indicate that compact feature release is not inherently privacy-preserving and that the released representation itself must be treated as sensitive.

We further analyze where this leakage originates. ANOVA-based feature responsibility analysis shows that the private attributes are encoded through different spectral and spatial feature groups. Gender leakage is most strongly associated with theta power, age with beta-relative power and asymmetry features, and subject identity with beta power and alpha–beta relationships. This attribute-specific structure limits the effectiveness of simple feature-level defenses. Uniform perturbation can damage task-relevant information together with private information, while fixed feature masking does not consistently suppress all private attributes.

To address this limitation, we present NeuroPriv, a learned representation framework for privacy-preserving wearable EEG feature release. Figure 1 shows the system pipeline. NeuroPriv operates between feature extraction and downstream inference. It maps compact EEG features and their pairwise interactions into a task-relevant latent representation, then uses adversarial learning to reduce information associated with gender, age, and subject identity before the representation is released. The transformed representation is used for the intended task, while external attackers are trained independently to measure residual privacy leakage. We make the following contributions:

- We demonstrate that compact EEG features leak gender, age, and subject identity even when raw EEG signals are not transmitted.
- We characterize the feature-level sources of privacy leakage and show that different private attributes are associated with different spectral and spatial feature groups.
- We evaluate Gaussian noise and feature masking as lightweight feature-level defenses and identify their limitations for multi-attribute privacy preservation.
- We design NeuroPriv, a learned representation framework that preserves task utility while suppressing gender, age, and subject identity information before feature release.

2 Related Work

EEG contains information beyond the signal component required by an intended application. Early work demonstrated that consumer EEG devices can be exploited as side channels to infer sensitive information from neural responses [12]. Subsequent studies showed that EEG also contains persistent subject-specific patterns that support biometric identification and cross-session re-identification [11, 14]. Cryptographic approaches use secure computation to prevent direct disclosure of EEG signals [2]. Federated and transfer learning approaches keep recordings local while sharing model parameters or pretrained models [10, 20]. Synthetic EEG generation supports downstream training without distributing the original patient recordings [17]. These approaches protect EEG during storage, sharing, or model training. They do not prevent sensitive attributes from being inferred from a feature vector or embedding that a system releases by design at inference time.

A parallel line of work partitions a network between a client device and a server, so the server never observes the raw input. Abuadbba et al. showed that intermediate activations of a 1D CNN over physiological time series remain highly correlated with the raw signal, which undermines the assumption that a split point alone confers privacy [1]. Aziz et al. instantiate split inference on multi-channel EEG headwear for seizure detection [4]. They introduce a Combined Privacy Metric that averages six normalized statistical, geometric, and information-theoretic quantities computed between the raw input and the intermediate activation, and they report that deeper convolutional split points yield higher combined privacy scores at increasing client-side cost.

Two properties of that formulation motivate our design. First, each component of the metric measures similarity between the released tensor and the raw signal. Low similarity does not imply that a private attribute is hard to recover, because an adversary is free to learn a nonlinear decision boundary in the released space. Second, each component is normalized by its maximum across the split points under evaluation, so the resulting score is relative to the configurations tested rather than an absolute leakage quantity. NeuroPriv therefore measures leakage operationally. We train independent external attackers on the released representation and report their accuracy against a chance baseline.

Recent work applies perturbation and representation learning to reduce private information while preserving EEG task utility [14]. Prior work extends adversarial perturbation to multiple private attributes, including identity, gender, and BCI experience [15]. Autoencoder-based anonymization and identity-removal networks similarly separate task-relevant information from subject-specific information [18, 19]. These studies demonstrate the value of learned privacy–utility optimization. They operate on raw EEG, reconstructed EEG, or task-specific signal perturbations.

Our method, **NeuroPriv**, addresses privacy at the compact-feature release boundary of a wearable EEG pipeline. We do not reconstruct or perturb the raw signal. We transform spectral and spatial EEG features into a latent representation before release. We quantify leakage for gender, age group, and subject identity using independently trained external attackers, identify the feature groups associated with each private attribute, and compare the learned representation against Gaussian perturbation and privacy-ranked feature masking. This lets us evaluate directly whether a released EEG representation preserves its intended utility while reducing several distinct forms of private-attribute inference.

3 Methods

3.1 Dataset and Features

We use the EEGMAT dataset [21], containing EEG recordings (36 people) collected before and during the performance of mental arithmetic tasks. Each subject contributes recordings under Baseline and Arithmetic conditions, and the utility target is to distinguish these two states. In addition to the task label, each sample is associated with subject-level private attributes: gender, age group, and subject identity.

Each EEG window is represented by a 17-dimensional handcrafted feature vector. The representation includes absolute band powers, relative band powers, spectral ratios, and spatial asymmetry measures. The frequency bands are delta, theta, alpha, beta, and gamma. The ratio features are theta/alpha, alpha/beta, theta/beta, and engagement. The asymmetry features are left–right alpha asymmetry, left–right beta asymmetry, and temporal–frontal alpha asymmetry.

3.2 Threat Model and Leakage Metrics

We consider a wearable EEG pipeline in which raw EEG remains on the device and only a derived representation is released for downstream inference. Depending on the system configuration, this released representation $\mathbf{z}$ may correspond to the original compact EEG features, a feature-level defended representation, or the latent representation produced by NeuroPriv. We treat $\mathbf{z}$ as the privacy boundary of the system: data upstream of $\mathbf{z}$ is protected by the device, whereas downstream services that operate on $\mathbf{z}$ may be untrusted or semi-trusted.

Each EEG window is represented by $\mathbf{x} \in \mathbb{R}^{17}$ and is associated with one utility label y_T and three private labels: gender y_G, age group y_A, and subject identity y_I. The intended utility task is Baseline-versus-Arithmetic classification. We assume that any service or third party with access to $\mathbf{z}$ may additionally attempt to infer these private attributes using auxiliary labeled data from the same population. The adversary does not observe the raw EEG, the parameters of the NeuroPriv encoder, or the training gradients.

In many application scenarios, these attributes are sensitive. Subject identity turns EEG into a biometric, enabling persistent re-identification and cross-service tracking even when other identifiers are removed [11, 14]. Demographic attributes such as gender and age support profiling and differential treatment, and can be combined with other data sources to link neural signals to medical or behavioral records [8]. We therefore treat successful prediction of y_G, y_A, or y_I from $\mathbf{z}$ as privacy leakage, even though the raw EEG never leaves the device and the representation is optimized for low-latency inference.

Utility is measured using task balanced accuracy:

$$U(\mathbf{z}) = \mathrm{BA}\big(y_T, h_T(\mathbf{z})\big), \tag{1}$$

where h_T is the downstream task classifier. Higher $U(\mathbf{z})$ indicates better preservation of task performance. Privacy leakage is measured using an attacker set $\mathcal{A} = \{\mathrm{SVM}, \mathrm{RF}, \mathrm{GB}\}$. For each private

attribute $p \in \{G, A, I\}$, we report the strongest attack performance:

$$P_p(\mathbf{z}) = \max_{a \in \mathcal{A}} \mathrm{BA}\big(y_p, a(\mathbf{z})\big). \tag{2}$$

Here, $P_p(\mathbf{z})$ captures the best balanced accuracy an attacker can achieve on attribute p given only $\mathbf{z}$ and auxiliary labeled data. Values near chance indicate limited recoverability of the corresponding attribute, whereas substantial gains over chance reflect significant leakage. The balanced-accuracy chance levels are 0.500 for the binary task and gender labels, $1/3 = 0.333$ for the three-class age label, and $1/36 = 0.028$ for closed-set subject identity. We apply this leakage evaluation to the original feature representation, to Gaussian-noise and feature-masking baselines, and to the representation produced by NeuroPriv.

To identify which EEG features contribute most strongly to leakage, we compute a one-way ANOVA F-value for each feature and private attribute. For feature j and private attribute p,

$$F_{j,p} = \frac{\sum_c n_c \left(\bar{x}_{j,c} - \bar{x}_j\right)^2 / (C-1)}{\sum_c \sum_{i \in c} \left(x_{i,j} - \bar{x}_{j,c}\right)^2 / (N-C)}. \tag{3}$$

Here, C is the number of classes, n_c is the number of samples in class c, $\bar{x}_{j,c}$ is the class-specific mean of feature j, and $\bar{x}_j$ is its overall mean. A larger F-value indicates stronger class separability and therefore greater potential leakage for the corresponding private attribute. We use these scores both to analyze feature-level leakage and to construct the feature-masking baseline.

3.3 Gaussian Noise

Gaussian noise is a coordinate-agnostic privacy baseline. It perturbs every standardized feature coordinate:

$$\tilde{\mathbf{x}} = \mathbf{x} + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}). \tag{4}$$

The defense uses no information about which features leak, so it applies equal perturbation to attribute-bearing and task-bearing coordinates. We sweep σ and audit task and identity at each level.

3.4 Feature Masking

Feature masking is a coordinate-targeted baseline. We rank EEG features using their combined contribution to gender, age, and subject identity leakage and zero the top-k privacy-ranked coordinates.

Let R_j denote the combined privacy-responsibility score of feature j across gender, age, and subject identity. The masking transformation is

$$M^{(k)}(\mathbf{x})_j = \begin{cases} 0, & \text{if } \mathrm{rank}(R_j) \le k, \\ x_j, & \text{otherwise.} \end{cases} \tag{5}$$

Because the features are standardized, setting a coordinate to zero replaces it with its training-set mean.

3.5 NeuroPriv Architecture

NeuroPriv transforms compact EEG features into a task-relevant representation with reduced private-attribute information. As shown in Figure 1, the framework consists of feature expansion, latent representation learning, and adversarial privacy suppression.

Given an EEG feature vector $\mathbf{x} \in \mathbb{R}^{17}$, we augment the original features with all pairwise interactions:

$$\phi(\mathbf{x}) = \left[\mathbf{x}; \{x_i x_j\}_{i<j}\right] \in \mathbb{R}^{153}. \tag{6}$$

The expanded representation is mapped to a compact latent representation through an encoder:

$$\mathbf{z} = f_\theta\big(\phi(\mathbf{x})\big). \tag{7}$$

A task head predicts the intended utility label from $\mathbf{z}$, while three privacy heads predict gender, age group, and subject identity. The privacy heads are connected to the encoder through gradient-reversal layers. During training, the task objective encourages $\mathbf{z}$ to retain task-relevant information, whereas the reversed privacy gradients discourage the encoder from preserving information useful for private-attribute inference.

The resulting optimization objective is

$$\min_{\theta, h_T} \max_{h_G, h_A, h_I} \mathcal{L}_T - \lambda_G \mathcal{L}_G - \lambda_A \mathcal{L}_A - \lambda_I \mathcal{L}_I, \tag{8}$$

where $\mathcal{L}_T$ is the task-classification loss and $\mathcal{L}_G$, $\mathcal{L}_A$, and $\mathcal{L}_I$ are the gender, age, and identity losses.

After training, the privacy heads are removed and only the transformed representation $\mathbf{z}$ is released for downstream task inference. Privacy is evaluated independently using external SVM, RF, and GB attackers trained on the released representation.

4 Evaluation

We evaluate four aspects of the proposed framework: (i) utility and privacy leakage in the original EEG feature representation, (ii) the EEG features most responsible for private-attribute leakage, (iii) Gaussian noise and feature masking as lightweight privacy baselines, and (iv) the privacy–utility performance of NeuroPriv.

Balanced accuracy (BA) is used for both utility and privacy evaluation. For the original feature representation, we report the performance of support vector machine (SVM), random forest (RF), and gradient boosting (GB) classifiers. For each private attribute, leakage is determined by the strongest evaluated attacker, following Eq. 2.The feature-level defense experiments and the NeuroPriv experiment were conducted using different data partitions. We therefore evaluate each method relative to the corresponding original-feature baseline from the same experimental protocol and do not directly compare absolute values across the two protocols.

Leakage in the original EEG representation. Table 1 reports task and private-attribute balanced accuracy for the original EEG feature representation. RF achieves the highest task balanced accuracy of 0.788. However, the same representation also supports gender, age, and subject-identity inference with balanced accuracies of 0.858, 0.789, and 0.692, respectively. Gender is inferred more accurately than the intended cognitive task. Subject-identity inference also substantially exceeds its closed-set chance level of $1/36 = 0.028$. These results show that compact EEG features preserve useful task information while simultaneously exposing multiple private attributes.

Feature responsibility for privacy leakage. Table 2 reports the highest-ranked EEG features according to their ANOVA F-values. The rankings differ across private attributes, indicating that privacy

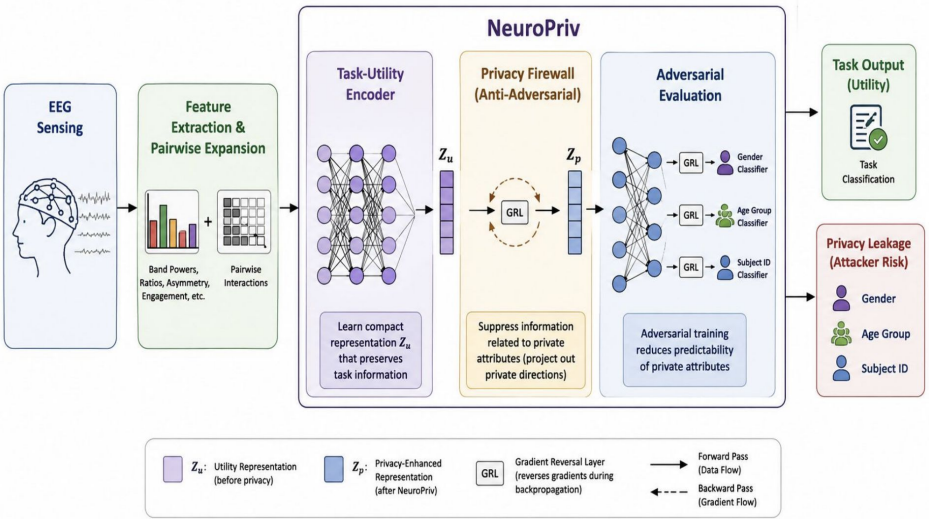


Figure 1: NeuroPriv architecture. EEG features and pairwise interactions are mapped to a task-relevant latent representation. Gradient-reversal adversaries suppress gender, age, and subject identity information before the representation is released for downstream inference.

Table 1: Balanced accuracy for the original EEG feature representation under each attacker.

Target	SVM	RF	GB
Task	0.639	**0.788**	0.763
Gender	0.755	0.845	**0.858**
Age	0.657	**0.789**	**0.789**
SubjectID	0.635	**0.692**	0.632

leakage is distributed across different spectral and spatial feature groups.

Gender is most strongly associated with theta absolute power ($F = 75.135$) and theta relative power ($F = 53.203$). Age is most strongly associated with beta relative power ($F = 42.683$) and temporal–frontal alpha asymmetry ($F = 41.877$). Subject identity is dominated by beta absolute power, which reaches an F-value of 116.644, followed by beta relative power and the alpha/beta ratio.

These results show that private information is attribute-specific rather than concentrated in a single common feature. Consequently, suppressing one highly ranked coordinate is unlikely to remove gender, age, and identity information simultaneously.

Gaussian noise. Table 3 reports task and private-attribute balanced accuracy under Gaussian perturbation. The $\sigma = 0$ configuration is the original-feature baseline for this experiment.

We select $\sigma = 0.30$ as the privacy-oriented operating point. At this setting, task balanced accuracy changes only from 0.608 to 0.602, while gender decreases from 0.691 to 0.633 and subject identity decreases from 0.718 to 0.594. However, age balanced accuracy increases from 0.322 to 0.415. The results indicate that isotropic perturbation affects private attributes differently and does not provide consistent multi-attribute suppression.

Table 2: Top ANOVA-ranked EEG features for private-attribute leakage.

Attribute	Rank	Feature	F-value
Gender	1	Theta absolute power	**75.135**
Gender	2	Theta relative power	53.203
Gender	3	Alpha/Beta ratio	41.027
Age	1	Beta relative power	**42.683**
Age	2	Temporal–frontal alpha asym.	41.877
Age	3	Left–right beta asym.	39.123
SubjectID	1	Beta absolute power	**116.644**
SubjectID	2	Beta relative power	49.659
SubjectID	3	Alpha/Beta ratio	46.664

Table 3: Balanced accuracy under Gaussian noise on standardized EEG features.

σ	Task ↑	Gender ↓	Age ↓	SubjectID ↓
0.00	0.608	0.691	0.322	0.718
0.03	0.585	0.715	0.363	0.710
0.05	0.611	0.688	0.339	0.703
0.10	0.605	0.669	0.351	0.683
0.20	0.617	0.659	0.409	0.642
0.30	**0.602**	**0.633**	**0.415**	**0.594**

Feature masking. Table 4 reports balanced accuracy after masking the highest-ranked privacy-related EEG features. The "None" configuration is the original-feature baseline for this experiment.

Table 4: Balanced accuracy under privacy-ranked feature masking. The selected operating point is shown in bold.

Masking	Task ↑	Gender ↓	Age ↓	SubjectID ↓
None	0.608	0.691	0.322	0.718
Top-1	0.617	0.674	0.360	0.715
Top-3	0.611	0.676	0.360	0.642
Top-5	0.588	0.683	0.377	0.536
Top-8	**0.608**	**0.664**	**0.363**	**0.518**

Top-8 is selected as the privacy-oriented masking configuration because it provides the largest subject-identity reduction while preserving task performance. SubjectID balanced accuracy decreases from 0.718 to 0.518, while task balanced accuracy remains unchanged at 0.608. Gender decreases modestly from 0.691 to 0.664, whereas age increases from 0.322 to 0.363.

Feature masking therefore provides a stronger reduction in identity leakage than Gaussian perturbation in this experimental setting, but its effect is not consistent across all private attributes.

4.1 Privacy–Utility Performance.

Table 5 compares NeuroPriv with the original EEG representation and the two feature-level baselines. Against the original-feature baseline from the same partition, NeuroPriv preserves task balanced accuracy at 0.781 vs 0.788, a reduction of 0.007 balanced-accuracy points.

At the same time, gender inference decreases from 0.858 to 0.563, age inference from 0.789 to 0.467, and subject-identity inference from 0.692 to 0.206, corresponding to reductions of 0.295, 0.322, and 0.486 balanced-accuracy points. Subject-identity inference remains above the closed-set chance level of 0.028, indicating that identity information is substantially reduced but not completely removed.

The Gaussian-noise and feature-masking rows were obtained under a different data partition and should be read against their own original-feature baseline (0.608 task, 0.691 gender, 0.322 age, 0.718 subject identity) rather than against the first row of Table 5. Relative to that baseline, neither lightweight defense reduces all three private attributes: both leave age inference higher than it started. NeuroPriv instead reduces gender, age, and subject identity within a single learned representation while maintaining task performance close to the original features.

Key takeaway. The results demonstrate that EEG feature release preserves substantial information about gender, age, and subject identity. Gaussian noise and feature masking reduce selected forms of leakage, but their effects vary across private attributes. NeuroPriv instead learns a single representation that preserves the intended task while jointly reducing all three evaluated private attributes.

5 Discussion

The results show that EEG feature release is not privacy-neutral. The original feature representation supports the intended task, but it also exposes demographic and identity information, and on this dataset it exposes gender more reliably than the task itself. This is a systems concern because compact feature release is widely treated as a safer alternative to raw-signal transmission.

The two baselines illustrate the privacy–utility tension directly. Gaussian noise and feature masking affect private attributes differently. Gaussian perturbation reduces gender and identity at larger noise levels but does not consistently reduce age inference. Feature masking provides stronger identity reduction while largely preserving task performance, but gender changes only modestly and age inference does not decrease. These results show that fixed feature-level defenses do not provide consistent multi-attribute privacy protection.Our NeuroPriv is more effective because it learns a new representation optimized jointly for task utility and multi-attribute suppression, and so acts on a subspace rather than on the original feature coordinates.

The implication for wearable EEG pipelines is that privacy should be treated as a property of the released representation rather than of the raw signal alone. Protection must operate before compact features leave the device.

6 Limitations and Future Work

Our evaluation has several limitations. First, we use a single dataset (EEGMAT) with 36 subjects in a controlled lab setting, which may not generalize to diverse populations or real-world deployment scenarios. Second, we evaluate closed-set subject identification over enrolled users; open-set or cross-session scenarios may present different privacy–utility tradeoffs. Third, our utility task is lab-based mental arithmetic, whereas real-world wearable EEG applications may involve different cognitive states, continuous monitoring, or multimodal sensing. Fourth, our attacker model assumes access to labeled auxiliary data from the same population; attackers with less or different side information may achieve weaker or stronger inference. Future work should validate NeuroPriv on diverse EEG datasets and deployment contexts, extend evaluation to open-set and cross-session identity scenarios, and explore additional privacy attributes such as health conditions, emotional states, or BCI experience. We also plan to deploy NeuroPriv on actual wearable devices to measure computational cost, latency, and energy consumption under real-time constraints, and to investigate whether the learned representation transfers effectively across tasks and populations.

7 Conclusion

This paper presented NeuroPriv, a learned representation framework for privacy-preserving wearable EEG feature release. We first showed that compact spectral and spatial EEG features are not privacy-neutral: the same representation used for cognitive-state inference also exposes gender, age, and subject identity. Feature-responsibility analysis further showed that this leakage is structured across different spectral and spatial feature groups. We evaluated Gaussian noise and feature masking as lightweight defenses and found that their effects are attribute-dependent, limiting their ability to provide consistent multi-attribute privacy protection. NeuroPriv addresses this limitation by learning a compact representation that preserves task-relevant information while adversarially suppressing private attributes before the representation is released. In validation, NeuroPriv reduces private-attribute inference by up

Table 5: Privacy–utility comparison across the evaluated methods. Each privacy value is the strongest of the SVM, RF, and GB attackers, following Eq. 2. For the original features, the strongest attacker is RF for Task (0.788), GB for Gender (0.858), RF/GB tied for Age (0.789), and RF for SubjectID (0.692).

Method	Task ↑	Gender ↓	Age ↓	SubjectID ↓
Original features	**0.788**	0.858	0.789	0.692
Gaussian noise ($\sigma = 0.30$)	0.602	0.633	0.415	0.594
Feature masking (Top-8)	0.608	0.664	**0.363**	0.518
NeuroPriv	0.781	**0.563**	0.467	**0.206**

to 0.486 balanced-accuracy points while maintaining task performance close to the original feature representation. These findings suggest that practical neuroprivacy for wearable EEG systems requires protection at the representation level rather than relying only on raw-signal avoidance, uniform perturbation, or fixed feature removal.

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